utility10_05-bijli-log

Bijli Log: manufacturing manual

> **Status: design only.** Nothing here has been built or tested as hardware. This manual tells you exactly what to make, buy, assemble, program and test, and how to accept or reject a unit; expect to correct it after the first prototype run (5 units).

Product summary. Plugs into a wall 5 V USB adapter (supplied). When mains disappears it keeps running from a backup cell, timestamps start and duration, keeps the last 32 cuts in a ring and shows count, total hours without power and the last cut.

Product line / price in the plan: Tier B · low-voltage electronics · prototype Rs 1220 · price Rs 1499. Status: designed and verified as far as a computer can; no prototype built, nothing field-tested.

1. What you are building: deliverables

Deliverable	Where	State
PCB: Gerbers, BOM, CPL, KiCad project	`utility10/pcb/release/bijlilog/`	**order-ready: every automatic check passes** (DRC errors 0, netlist match True)
Firmware (`boards/bijli_log.hex`)	`utility10/firmware/release/4MHz/boards/bijli_log.hex`, source `utility10/firmware/src/main_board.cpp` (seeti: `seeti_prod.cpp`), drivers `board_drivers.h`, engine `utility10/firmware/lib/utilcore.c`	engine host-tested (115 checks); board firmware compiles but is **not yet run on hardware** (see section 6)
Enclosure STL (prototype shell)	`utility10/05-bijli-log/enclosure_body.stl`, `enclosure_lid.stl` and the board's `4_enclosure/`	prototype shell only
Mechanical / bought-in parts	section 3	listed
Test sheet	`manuals/templates/eol_test_sheet.md`	template

2. Before you start

- Read `manuals/STANDARD_PROCEDURES.md` (sections A-F, J, K).
- Tools: soldering iron, flux, multimeter, USBasp programmer + avrdude, calipers, 3D printer or print service, bench supply with current limit.
- Power source for this product: **5 V USB charger + 2 x AAA backup**.

3. Bill of materials

3a. Circuit board parts (28 parts, 3 test points)

SMD parts marked `SMT (JLC)` are assembled by the fab from the JLC BOM; parts marked `hand/THT` are soldered by you. Every LCSC number was checked against an LCSC/JLCPCB listing on 2026-11-01; re-check stock on the day you order.

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ATmega328P-AU	TQFP-32_7x7mm_P0.8mm	ATMEGA328P-AU	C14877	SMT (JLC)
C1	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C2	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C3	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
R7	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
J2	ISP	PinHeader_2x03_P2.54mm_Vertical	ZHOURI 2.54-2*3	C5116479	hand/THT
J1	USB-C 5V	USB_C_Receptacle_HRO_TYPE-C-31-M-12	TYPE-C-31-M-12	C165948	SMT (JLC)
F1	PTC 500mA	Fuse_1206_3216Metric	Littelfuse 1206L050YR (6 V)	C163512	SMT (JLC)
D3	SMAJ5.0A	D_SMA	Littelfuse SMAJ5.0A	C83329	SMT (JLC)
R1	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
R2	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
C5	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
D1	Green	LED_0603_1608Metric	Everlight 19-217/GHC-YR1S2/3T (LCSC showed it out of stock on the search day: check)	C72043	SMT (JLC)
R5	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
U3	AP2112K-3.3	SOT-23-5	Diodes AP2112K-3.3TRG1 (from the LCSC datasheet link)	C51118	SMT (JLC)
C6	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
C7	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
D5	BAT54WS	D_SOD-323	BORN BAT54WS (300 mA listing)	C5148476	SMT (JLC)
D6	BAT54WS	D_SOD-323	BORN BAT54WS (300 mA listing)	C5148476	SMT (JLC)
J6	BATT 2xAAA backup	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
C8	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
R12	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
R13	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
J4	OLED 0.91in I2C	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
J5	RTC DS3231 module	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
R10	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
R11	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
SW2	Scroll / clear	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)

3b. Bought-in and mechanical items (estimates for about 2,000 units; replace with quotes)

Item	Rs
MCU	45
RTC module	60
Display 0.91 in OLED	90
Backup cell + charger	40
USB adapter (BIS-registered, bought in)	90
PCB	30
Enclosure	50
Packaging	30
Assembly + test	40

4. Step 1: fabricate the PCB

Upload `utility10/pcb/release/bijlilog/1_fabrication/bijli_log_gerbers.zip` with the settings in STANDARD_PROCEDURES section A. Finish: HASL lead-free. Board size and layer list are in `utility10/pcb/release/bijlilog/README.md`.

5. Step 2: assemble the board

1. Order SMT assembly as in STANDARD_PROCEDURES section B (or hand-solder the SMD parts in section D order).
2. Solder the through-hole parts listed `hand/THT` in 3a.
3. Clean, inspect under magnification, and compare with `2_assembly/*_assembly_top.pdf`.

6. Step 3: program the microcontroller

1. Connect a USBasp to the 6-pin ISP header J2 (pin 1 = square pad) with the board powered (5 V USB charger + 2 x AAA backup).
2. Set fuses: `avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFE:m` (internal 8 MHz RC, CKDIV8 off, brown-out 1.8 V; the firmware then divides the clock to 4 MHz (2 x AAA

supply may fall to 2.0 V)).
3. Flash and verify: `avrdude -p m328p -c usbasp -U flash:w:utility10/firmware/release/4MHz/boards/bijli_log.hex:i` then `-U flash:v:utility10/firmware/release/4MHz/boards/bijli_log.hex:i`.
4. Remove the programmer, power-cycle the board, and confirm the start-up behaviour (section 9).

If you change the supply type (for example USB instead of batteries) use the hex built for that clock (`4MHz` for 2 x AAA, `8MHz` for every other tool).

Firmware pin map (generated from the PCB spec, so firmware and board agree):

Arduino pin	ATmega pin	Net
D7	11	LED
A0	23	MAINS
A4	27	SDA
A5	28	SCL
D2	32	BTN

What is verified: the control logic (`utilcore.c`) passes 115 host tests. **What is not:** the display, OLED, RTC and HX711 drivers and the button flows in `main_board.cpp` were written from datasheets and only compile-checked. Budget bring-up time for them on the first prototype.

7. Step 4: electrical bring-up

1. Plug USB: +3V3 = 3.3 V, VSYS about 3.0 V, MAINS (TP3) about 2.5 V.
2. Unplug: MAINS goes low, unit stays alive on the battery, cut logged; replug: duration shown.

8. Step 5: sensors, cables and mechanical assembly

- **Modules.** OLED 0.91 in I2C module on J4 and DS3231 RTC module on J5 (GND, +V, SDA, SCL order is printed on the board). Fit a CR2032/AAA backup as per the README; use a certified, BIS-registered 5 V adapter: the unit itself is low voltage.

9. Step 6: enclosure and final assembly

1. Print the shell (STANDARD_PROCEDURES section G). Enclosure: `enclosure_body.stl` / `enclosure_lid.stl` (70×50×24 mm, 2.4 mm walls, watertight): a simple prototype shell; production tooling needs a proper design.
2. Place the board on the standoffs (4 mm), fasten with M2.5 screws (6 mm), route cables through the cut-outs, close the lid.
3. Label: serial number, rev, battery type/polarity, warnings.

10. Step 7: calibration

- Set the RTC time once at the factory (and again by the user after a battery replacement). Check the RTC drift over 24 h (module spec about +-2 ppm for DS3231; clone modules vary).

11. Step 8: end-of-line functional test (every unit)

1. Remove 5 V for 10 s: one cut of 10 s +-1 s is logged; restore power.

2. Pull the cable again twice: the count increases and the last-cut screen updates.
3. Check the log survives a full power removal (backup cell holds the RTC).

Record results on `manuals/templates/eol_test_sheet.md`. Accept only if every line passes.

12. Packaging, documents, compliance

- In the box: unit, batteries or note on the required adapter, mounting accessories, quick-start card (`manuals/templates/quick_start_card.md`), warranty card.
- Compliance items to verify before sale: Battery safety, BIS requirement for any bought-in adapter, EMC, e-waste EPR, Legal Metrology labelling; no safety claims..
- Labelling: STANDARD_PROCEDURES K.

13. Troubleshooting (product specific)

Symptom	Cause / action
Time resets after power loss	RTC battery missing or dead
OLED blank	wrong module pinout: some modules swap GND and VCC

General issues: STANDARD_PROCEDURES section L.

14. Known limits and risks

Host tests: a 1-hour cut is logged with the right duration, 1 s flickers are ignored (2 s debounce), 40 cuts keep the last 32 and total outage adds up. Not verified: real clock drift (needs an RTC), backup cell life.

Adapter compliance, a crowded smart-plug market (but none give a simple log), feature creep.

Nothing in this manual replaces testing with real users, a design review by a PCB engineer, and the compliance checks above.